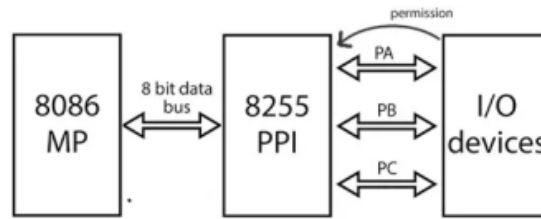


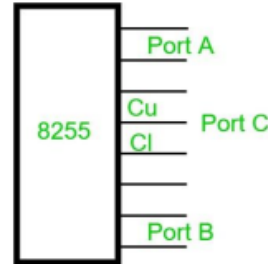
09. 8255(I/O Controller)

46.

- The 8255 is a general purpose programmable IO device. It is designed to interface the CPU with some external devices like ADC, DAC, keyboard etc.



- There are three 8-bit bi-directional IO ports. The Port-A, Port-B, and Port-C. These ports are assigned as input or output port.



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The block diagram of 8255 is like this:

Port A= 8 I/O lines= PA7-PA0

Port B= 8 I/O lines= PB7-PB0

Port C= 4 I/O lines= PC7-PC4

= Upper lines of Port C

Port C= 4 I/O lines= PC3-PC0

= Lower lines of Port C

I/O= Input/Output

CRW= Control Register Word

= Store 8-bit data= D7-D0

RD_b= Read Bubble line= Active with 0

WR_b= Write Bubble line= Active with 0

CS_b= Chip Select Bubble line= Active with 0

A0,A1= Address lines

For READ operation: RD_b=0, WR_b=1, CS_b=0

For WRITE operation: RD_b=1, WR_b=0, CS_b=0

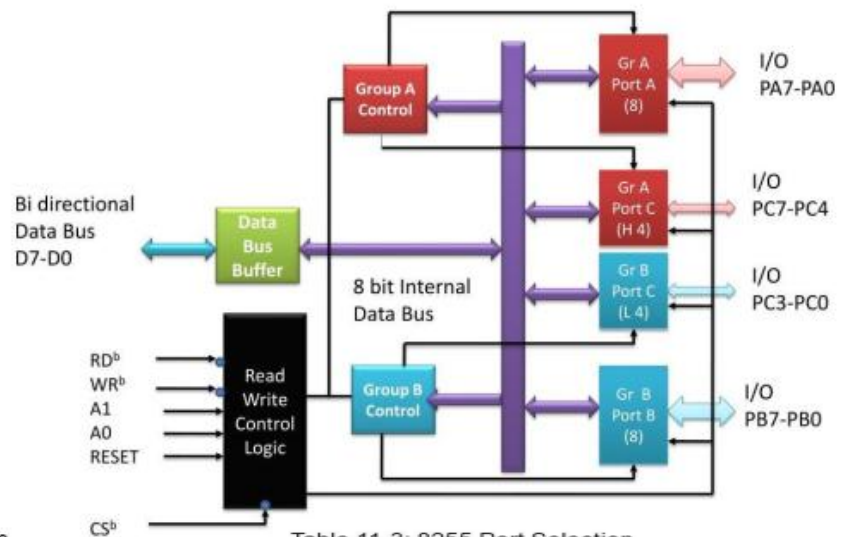
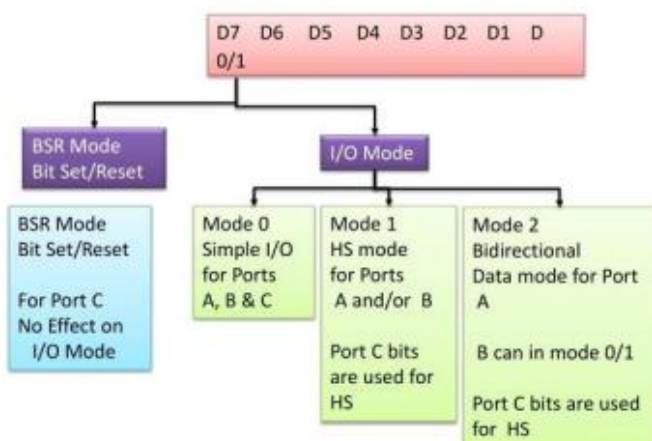


Table 11-3: 8255 Port Selection

CS	A1	A0	Selects
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	x	x	8255 is not selected

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Ports & Modes in 8255



- There are two different modes of 8255. These modes are:

1. Bit Set Reset (BSR) Mode- This mode is used to set or reset the bits of the Port-C only. For BSR mode always D7 will be 0.

2. Input/ Output Mode- This mode is selected when the D7 bit of the control register is 1.

- This mode has also three different modes. These modes are Mode 0 and Mode 1 and Mode 2.

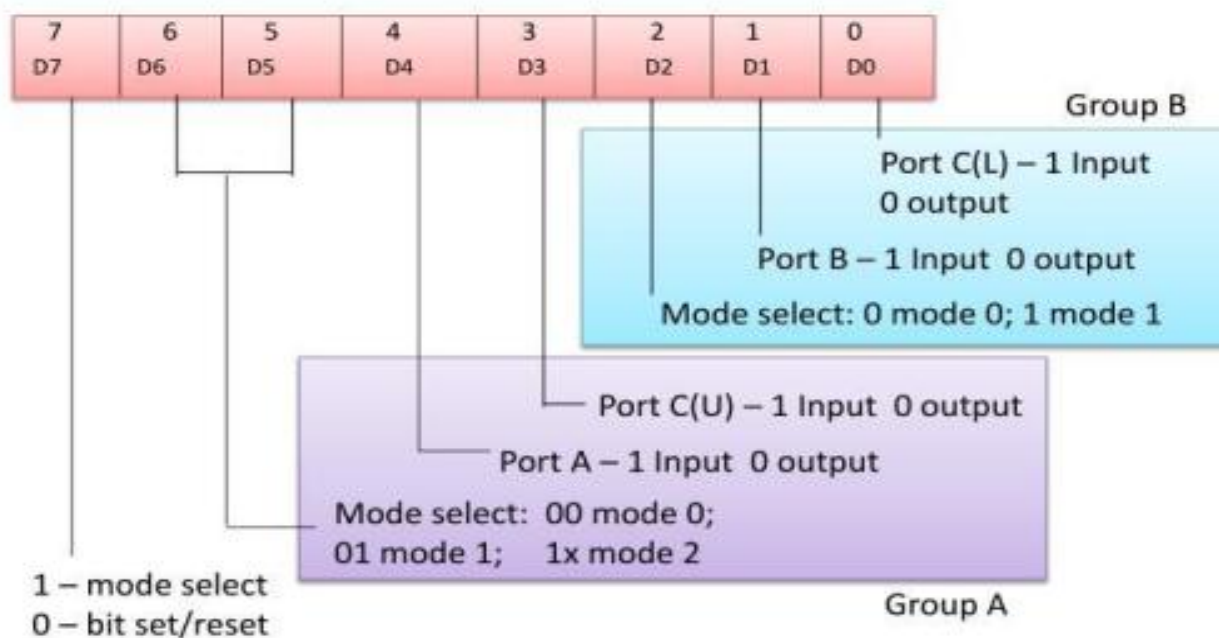
- **Mode 0** – Simple or basic I/O Mode- In this mode all of the ports A, B and C can be used as input or output mode. The outputs are latched, but inputs are not latched. This mode has interrupt handling capability.
- **Mode 1** – Handshake or Strobed I/O- In this mode the Port A and Port B can be used as input or output ports, the port C are used for handshaking. In this mode the inputs and outputs are latched. This mode also has the interrupt handling capability, and signal control to match the speed of CPU and IO devices.
- **Mode 2** – Bidirectional I/O- In this mode only Port A can work, and port B can either be in mode 0 or mode 1, and the port C are used for handshaking. In this mode the inputs and outputs are latched.

****latch(a temporary storage register)**

****Latched Outputs/ Inputs:** When a port is configured as an output and is "latched," it means that the data **written/ read to/ from** the port is stored in a latch and remains stable at the **output/input** pins until new data is **written/read** to the port.

49.

Ports & Modes in 8255 : Control register



Bits	Function
D7	1 for IO mode and 0 for BSR mode
D6 & D5	These are used to set port A mode. for 00, it is m0 mode, for 01, it is m1 mode and 10 or 11, it is m2 mode.
D4	1 when port A is taking input, 0 when port A is sending output.
D3	1 when higher nibble of port C is taking input, and 0 when higher nibble is sending output.
D2	It tells the mode of Port B. For 0, it is m0 mode, and for 1, it is m1 mode.
D1	1 when port B is taking input, 0 when port B is sending output.
D0	1 when lower nibble of port C is taking input, and 0 when lower nibble is sending output.